

1. (Fall 2024 #3) Find the derivative of the following functions. You do not need to simplify your answer.

(a) $f(x) = \tan\left(\frac{e^{\sqrt{x}}}{\ln x}\right)$

$$\frac{d}{dx}[\tan(u)] = \frac{d}{dx}\left[\frac{\sin(u)}{\cos(u)}\right] = \frac{\cos^2(u) + \sin^2(u)}{\cos^2(u)} = \frac{1}{\cos^2(u)} = \sec^2(u)$$

Let $u = \frac{e^{\sqrt{x}}}{\ln x}$. By chain rule: $f'(x) = \sec^2(u(x)) \cdot u'(x)$

By quotient rule: $u'(x) = \frac{\left(\frac{1}{2\sqrt{x}}e^{\sqrt{x}}\right)\ln x - e^{\sqrt{x}} \cdot \frac{1}{x}}{(\ln x)^2}$

So $f'(x) = \sec^2\left(\frac{e^{\sqrt{x}}}{\ln x}\right) \cdot \frac{e^{\sqrt{x}} \left(\frac{\ln x}{2\sqrt{x}} - \frac{1}{x}\right)}{(\ln x)^2}$

(b) $f(x) = (\sin x)^{\cos^2 x}$

Taking the natural log: $\ln(f(x)) = \cos^2 x \cdot \ln(\sin(x))$

Thus: $\frac{f'(x)}{f(x)} = 2\cos x \cdot (-\sin x) \ln(\sin(x)) + \cos^2 x \cdot \frac{\cos x}{\sin x}$

Multiplying by $f(x) = (\sin x)^{\cos^2 x}$:

$$f'(x) = (\sin x)^{\cos^2 x} \left(-2\cos x \cdot \sin x \cdot \ln(\sin x) + \frac{\cos^3 x}{\sin x} \right)$$

2. (Spring 2024, #3) Find the derivatives of the functions given below. You do not need to simplify the final form of your derivative.

(a)

$$g(x) = \frac{x}{(x^2 + \tan(x))^5}$$

Let $f(x) = x$, $h(x) = (x^2 + \tan(x))^5$. By quotient rule: $g'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$

$$f(x) = x \Rightarrow f'(x) = 1$$

$$h(x) = (x^2 + \tan(x))^5 \xrightarrow{\text{by chain rule}} h'(x) = 5(x^2 + \tan(x))^4 (2x + \sec^2(x))$$

I used the fact that $\tan'(x) = \sec^2(x)$ which we have already shown in Q1.

$$\text{Thus: } g'(x) = \frac{(1)(x^2 + \tan(x))^5 - x \cdot 5(x^2 + \tan(x))^4 (2x + \sec^2(x))}{(x^2 + \tan(x))^{10}}$$

(b)

$$f(x) = x^2 \cdot \cos(\ln(x^3 - 1) + 2)$$

Let $g(x) = x^2$, $h(x) = \cos(\ln(x^3 - 1) + 2)$. By product rule: $f'(x) = g'(x)h(x) + g(x)h'(x)$

$$g(x) = x^2 \Rightarrow g'(x) = 2x$$

$$h(x) = \cos(\ln(x^3 - 1) + 2) \xrightarrow{\text{by chain rule}} h'(x) = -\sin(\ln(x^3 - 1) + 2) \cdot \frac{3x^2}{x^3 - 1}$$

$$\text{Since } \frac{d}{dx} [\ln(x^3 - 1) + 2] = \frac{\frac{d}{dx} [x^3 - 1]}{x^3 - 1} = \frac{3x^2}{x^3 - 1}.$$

Thus:

$$f'(x) = 2x \cdot \cos(\ln(x^3 - 1) + 2) + x^2 \cdot \left[-\sin(\ln(x^3 - 1) + 2) \cdot \frac{3x^2}{x^3 - 1} \right]$$

3. (Spring 2023, #1) Find the derivatives of the following functions. You do not need to simplify your answers.

(a) $f(x) = x^{e^x}$

Taking the natural log: $\ln(f(x)) = e^x \ln(x)$

Differentiate both sides: $\frac{f'(x)}{f(x)} = e^x \ln(x) + \frac{e^x}{x}$

Multiply both sides by $f(x)$: $f'(x) = x^{e^x} \left(e^x \ln(x) + \frac{e^x}{x} \right)$

(b) $h(x) = \frac{\cos(x^3 + 2x)\sqrt{x}}{\tan(x)}$

Let $f(x) = \cos(x^3 + 2x)\sqrt{x}$, $g(x) = \tan(x)$. By quotient rule: $h'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}$

$f(x) = \cos(x^3 + 2x)\sqrt{x}$

By Product rule $\implies f'(x) = [-\sin(x^3 + 2x)(3x^2 + 2)\sqrt{x}] + [\cos(x^3 + 2x) \cdot \frac{1}{2\sqrt{x}}]$

$g(x) = \tan(x) \implies g'(x) = \sec^2(x)$

$\implies h'(x) = \frac{\left[(-\sin(x^3 + 2x)(3x^2 + 2)\sqrt{x} + \cos(x^3 + 2x)\frac{1}{2\sqrt{x}}) \tan(x) \right] - (\cos(x^3 + 2x)\sqrt{x} \cdot \sec^2(x))}{[\tan(x)]^2}$

4. (Fall 2022, #2)

(a) Find every value of k such that the following function is continuous everywhere:

$$f(x) = \begin{cases} kx - 4 & x \leq 2 \\ x^3 & x > 2 \end{cases}$$

$$\lim_{x \rightarrow 2^-} f(x) = k(2) - 4 = 2k - 4$$

$$\lim_{x \rightarrow 2^+} f(x) = 2^3 = 8$$

$$f(2) = k(2) - 4 = 2k - 4$$

$$\text{For continuity we must have: } 2k - 4 = 8 \Rightarrow 2k = 12 \Rightarrow \boxed{k = 6}$$

(b) Let

$$g(x) = e^{\cos(\log_7(x))}$$

Use derivative rules to find $g'(x)$.

$$\text{Let } u(x) = \cos(\log_7(x)). \text{ By chain rule } g'(x) = e^{u(x)} \cdot u'(x)$$

$$u(x) = \cos(\log_7(x)) \xrightarrow{\text{By chain rule}} u'(x) = -\sin(\log_7(x)) \frac{d}{dx} [\log_7 x]$$

$$\text{We recall: } \frac{d}{dx} [\log_a x] = \frac{1}{x \ln(a)}.$$

$$\text{Thus: } u'(x) = -\sin(\log_7(x)) \cdot \frac{1}{x \ln 7} \Rightarrow g'(x) = e^{\cos(\log_7(x))} \left(-\sin(\log_7(x)) \cdot \frac{1}{x \ln(7)} \right)$$

(c) Let

$$g(x) = \frac{1}{x+1}$$

Find $g'(1)$ using the limit definition of the derivative.

$$\begin{aligned} g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \Rightarrow g'(1) = \lim_{h \rightarrow 0} \frac{g(1+h) - g(1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{1}{2+h} - \frac{1}{2}}{h} = \lim_{h \rightarrow 0} \frac{\frac{2 - (2+h)}{2(2+h)}}{h} = \lim_{h \rightarrow 0} \frac{\frac{-h}{2(2+h)}}{h} \\ &= \lim_{h \rightarrow 0} \frac{-1}{2(2+h)} = -\frac{1}{4} \end{aligned}$$

5. (Spring 2022, #2) Compute the derivatives of the following functions. Do not simplify your answer.

(a) $f(x) = \frac{e^x + e^{-x}}{e^x - e^{-x}}$

Let $g(x) = e^x + e^{-x}$, $h(x) = e^x - e^{-x}$. By quotient rule: $f'(x) = \frac{g'(x)h(x) - h'(x)g(x)}{h^2(x)}$

$$g(x) = e^x + e^{-x} \Rightarrow g'(x) = e^x - e^{-x}$$

$$h(x) = e^x - e^{-x} \Rightarrow h'(x) = e^x + e^{-x}$$

Thus:

$$f'(x) = \frac{(e^x - e^{-x})(e^x - e^{-x}) - (e^x + e^{-x})(e^x + e^{-x})}{(e^x - e^{-x})^2} = \frac{(e^x - e^{-x})^2 - (e^x + e^{-x})^2}{(e^x - e^{-x})^2}$$

(b) $y = \tan(x)^{2x}$

Taking natural log: $\ln y = 2x \cdot \ln(\tan(x))$

Differentiate both sides: $\frac{y'}{y} = 2 \ln(\tan(x)) + 2x \cdot \frac{\sec^2(x)}{\tan(x)}$

Multiply by $y = \tan(x)^{2x}$:

$$y' = \tan(x)^{2x} \left(2 \ln(\tan(x)) + \frac{2x \cdot \sec^2(x)}{\tan(x)} \right)$$